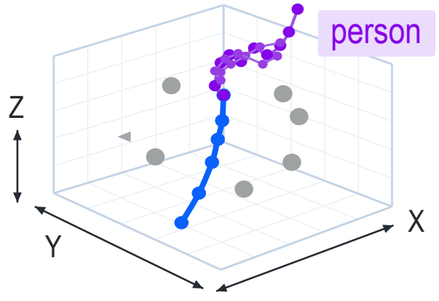
1 Sparse 4D object tracks (example)



3D tracks in world frame (same object)

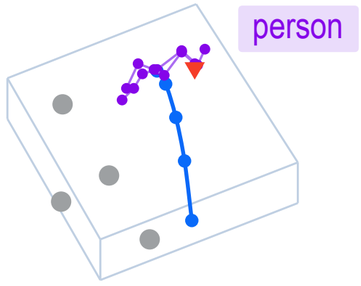


video clip (~34 frames)

per-object 3D keypoint tracks

2 Scene normalization

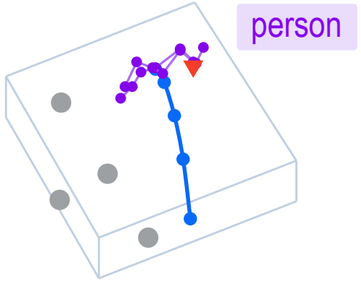
shared center + scale



shared center + scale
metric recovery retained

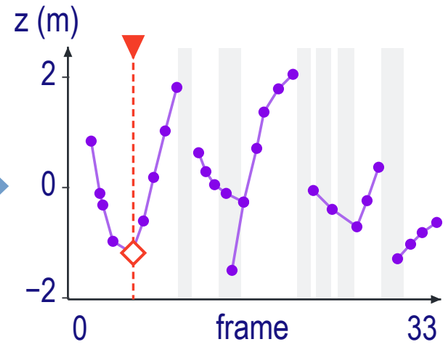
compute_scene_norm

normalize to a canonical world frame



shared center + scale
metric recovery retained

3 Split by visibility + events



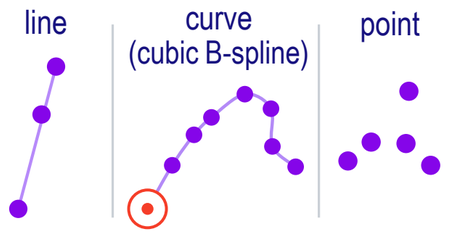
visible track

reversed

impact

occlusion gap (not visible)

4 Cubic B-spline / line / point fitting

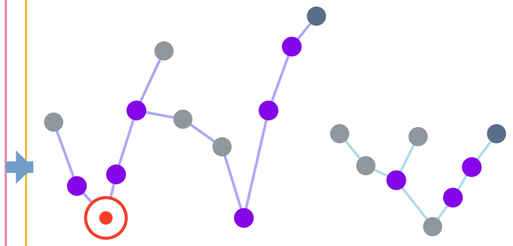


max error ≤ 0.02 × bbox diag

rigid objects & camera supported



5 On-curve nodes → tokens



on-curve node (keep)

other samples (discard)

event node (reversal/impact)

one token per spline coefficient (on-curve node)

6 Serialize into prompt (one trajectory stream per object)

tokenize_scene

Trajectory sequence in prompt

[scene] … [the person] [part 1] <traj> … <traj> … <traj> … [the camera] <traj> … <traj> …

One trajectory token record (for an on-curve node)

discrete fields (categorical)	role node	knot reversal	kind —	otype —	inst 5	part 2	kp 6
continuous fields (normalized)	xyz (0.61, 0.72, 0.48)			u 0.1	t 0.3		

7 Project to 4096-d embeddings

embeddings + linear projections

concat

Linear → 4096

LayerNorm

residual MLP

LayerNorm

…